Test Report No. 2024 – 0613

Agricultural Machinery – Four-Wheel Tractor
KUBOTA L3228DT

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Test Report Number	2024 – 0613
Date of Issuance	February 5, 2025
Valid Until	February 5, 2030
Agricultural Machinery	Four-Wheel Tractor
Type	Four-wheel drive, Three-cylinder, Naturally-aspirated, 8F + 8R transmission system, Mechanical steering system
Brand	KUBOTA
Model	L3228DT
Test Requested By	KUBOTA PHILIPPINES INC. 232 Quirino Highway, Baesa, 11096, Quezon City

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS PABES 301:2020	AMTEC Test
Rated Maximum Power, kW	23.5[a]	23.5[a]
Rated Engine Shaft Speed, rpm	2700[a]	2700[a]
Maximum PTO Power, kW	20.4[a]	21.32
Percentage PTO Power of the Engine Power, %, minimum	85.0	90.73
Specific Fuel Consumption, g/kW-h, maximum	350	265.72[b]
At Maximum PTO Power		
Engine Speed, rpm	ND	2720
PTO Shaft Speed, rpm	ND	600
PTO Torque, N-m	ND	339.83
Fuel Consumption, L/h	ND	6.75
At 540 rpm PTO Shaft Speed		
Engine Speed, rpm	ND	2365
PTO Torque, N-m	ND	361.50
PTO Power, kW	ND	20.43
Fuel Consumption, L/h	ND	6.07
Specific Fuel Consumption, g/kW-h	ND	249.57

[a] The data were obtained from the operator's manual submitted by the test applicant.
[b] Specific fuel consumption at maximum PTO power.
Note: ND – No Data

SUMMARY (Cont'n)

Performance Criteria	Requirement as per PNS/BAFS PABES 301:2020	AMTEC Test
Main Gear	1, 2, 3, 4, N	1, 2, 3, 4, N
Range Gear	High, Low, Neutral	High, Low, Neutral
Shuttle Gear	Forward, Reverse, Neutral	Forward, Reverse, Neutral
Slowest Forward Speed, kph	1.70	1.83
Fastest Forward Speed, kph	19.70	20.19
Radius of Turning Circle, mm (Without Brakes)		
Right Turn Left Turn	ND ND	3330 3295
Position and/or Draft Implement Controls	Shall be provided	Provided; Position controls only
Three-Point Linkage		
Category based on PTO power at rated rotational frequency of engine	1N: up to 35 kW 1: up to 48 kW	1N, 1
Category based on dimensions of tractor linkage points	1N/1: d_1: 19.3+0.2 d_2: 22.4+0.25	1N/1
Power-Take-Off Shaft	540 rpm shall have 6 straight splines or 100 rpm shall have 21 involute splines	540 rpm (6 straight splines)
Drawbar Category based on PTO power	1: PTO power ≤48 2: PTO power ≤115 3: PTO power ≤185 4: PTO power ≤200 5: PTO power ≤500	0, 1, 2, 3, 4, 5
Drawbar Power, kW	ND[a]	18.34[b]
Hydraulic Lift Force Capacity, kN	5.68[c]	6.27[d]
Wheel Tread Adjustment	Shall be provided	Provided
PTO Casing/Cover	Shall be provided	Provided
Roll-Over-Protective Structure (ROPS)	Shall be provided	Provided
Protective Seat Belt	Shall be provided	Provided

[a] Data were obtained from the operator's manual submitted by the test applicant.
[b] 86% of maximum PTO power.
[c] Computed as per Table 8 of PNS/ BAFS/ PABES 301:2020, requirement if drawbar power is equal to 65 kW or below.
[d] Without booster cylinder. 610 mm from the lower link endpoint.
Note: ND – No Data

OBSERVATIONS

1. Operator's manual	
a. Language	English
b. Operator's manual have its own part number and date of issue	Not provided
c. Section that provides information that will enable the operator to locate and identify the whereabouts of serial numbers and/or codes of the major components of the machine	Provided
d. Introduction is included in the OM stressing the importance of the information given in the manual.	Provided
e. Attention is drawn to the use of safety alert symbol to highlight information about potential dangers to the user.	Provided
f. Content list is provided to identify the main categories of information in the manual and where they can be found. Location: Right-hand page	Provided
g. Safety signs that appear on the machinery are reproduced in legible size in the operator's manual.	Provided
i. Location of each safety sign on the machine	Provided
ii. Instructions on the need to keep safety signs clear and visible on the equipment	Provided
iii. Instructions to replace safety signs if they are missing or illegible	Provided
iv. Instructions that new equipment components installed during repair shall include the current safety signs specified by the manufacturer and shall be affixed to the replacement component	Provided
v. Instructions on how to obtain replacement safety signs	Provided

OBSERVATIONS (Cont'n)

h. Operating information with logical instructions for effective operation of the machinery.	Provided
i. Authorized accessories and attachments and its affect in safety, operation and maintenance	Provided
j. Maintenance instructions and schedules are provided	Provided
k. Instructions and information regarding the proper storage of the machinery including precautions to be taken and any tools or special equipment required to prepare the machinery for storage.	Provided
l. Technical information and instructions for handling, reception, transportation, assembly, installation and initial set-up of the machinery	Provided
m. Relevant dimensions and technical data necessary to assist the operator achieve a higher standard of operational performance and reliability (Specifications of the machinery)	Provided
n. Action to be taken on the completion of the useful life of the machine or its parts, with instructions on dismantling and disposal.	Provided
o. Parts list is included within the operator's manual (if there is no separate parts list or catalogue exists)	Provided
p. Manuals are available in the language of each country where the machine is sold (Filipino or English)	English version is provided.
q. Consistent forms of language, spelling, numbering, symbols, etc. are used throughout the manual	The language, spelling, numbering, symbols, etc. are used throughout the manual.
r. All numbers are written in Arabic numerals	All numbers were written in Arabic numerals.
s. Title and number are provided for each table.	Provided
t. Illustrated parts catalogue and repair manual	Provided
u. Specific parts covered by warranty	Not provided

OBSERVATIONS (Cont’n)

2. Previous test reports of similar brand and model	None
3. Manufacturer	KUBOTA CORPORATION Thailand
4. Serial number	KBULRADRCRTE00003
5. Failures or abnormalities that may be observed on the machine or its components during and after the operation.	No failure or abnormalities were observed on the tractor or its component parts during and after the operation.
6. All tractors shall be equipped with Roll-Over Protective Structures (ROPS) and seat belts.	Provided
7. Seat shall be provided which will adequately support the operator in all working and operating conditions. Adequate and comfortable support and protection for the feet shall also be provided. Seats and ROPS shall conform to the requirements specified in PAES 139 and 140: 2004.	a. The tractor seat is adjustable and is provided with a seat belt. b. The tractor is provided with two post ROPS with a dimension of 100 mm × 54 mm. c. Canopy was attached to the posts through bolts and nuts. d. Conformity of the tractor seat and the ROPS to PAES 139 and 140 were not verified due to limited technical capability.
8. When the PTO is in use, a cover or casing that protects the sides of the PTO shaft shall be fitted. An additional non-rotating casing shall also be provided when the PTO is not in use. This casing shall enclose the PTO shaft completely and be fixed to the tractor body.	There is a provision for PTO cover that protects the sides of the PTO shaft. However, a non-rotating casing is not provided for the PTO during verification.
9. The engine of the tractor shall be equipped with cooling system suitable for tropical operations	The four-wheel tractor is provided with a water-cooling system.

OBSERVATIONS (Cont'n)

10. Mechanism that minimizes/ reduces vibration shall be provided	It was not verified if the machine's vibration was minimized/reduced due to limited technical capabilities in quantifying and comparing vibration levels during the test operation.
11. The four-wheel tractor shall be free from manufacturing defects that may be detrimental to its operation	The four-wheel tractor is free from any manufacturing defects.
12. Any uncoated metallic surface shall be free from rust and shall be painted properly	The four-wheel tractor is properly painted and is free from rust.
13. Appropriate labels, safety symbols, and warnings shall be provided appropriately.	Labels, safety, symbols, and warnings are provided (Figure 6 and 7).
14. All hot surfaces that may come in contact with the operator shall be provided with heat shield or protective cover.	Provided
15. Attachment of ballast or other weights shall be provided, as necessary.	Provided
16. The four-wheel tractor shall be free from sharp edges and surfaces that may injure the operator. Warning notices shall be provided in accordance with PAES 101:2000.	a. The tractor is free from sharp edges and rough surfaces. b. Warning notices are provided in the form of stickers and are affixed to the parts that may cause danger to the operator. The warning notices are visible to the operator and indicate the particular danger that the operator is at risk of (Figure 6).
17. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	November 8 & 18, 2024
Location of Test	Roberto C. Bautista Hall, UPLB, College, Los Baños, Laguna
Items Tested	Operator's Manual, PTO Load Test, Hydraulic Lift, Traveling Speed and Radius of Turning Circle/Area

METHODS OF TEST

The Four-Wheel Tractor was tested based on the PNS/BAFS PABES 302: 2020 – Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

Figure 1. The different parts of the KUBOTA L3228DT Four-Wheel Tractor include the following: (a) Canopy, (b) Operator's seat, (c) Steering wheel, (d) Exhaust, (e) Rear wheel, (f) Front Ballast, and (g) Front wheel, (h) Roll-over Protective Structures (ROPS), (i) Upper link hitch point, (j) Lower link hitch points, (k) PTO shaft, (k) and (l) Drawbar.

DESCRIPTION OF THE MACHINE (Cont'n)

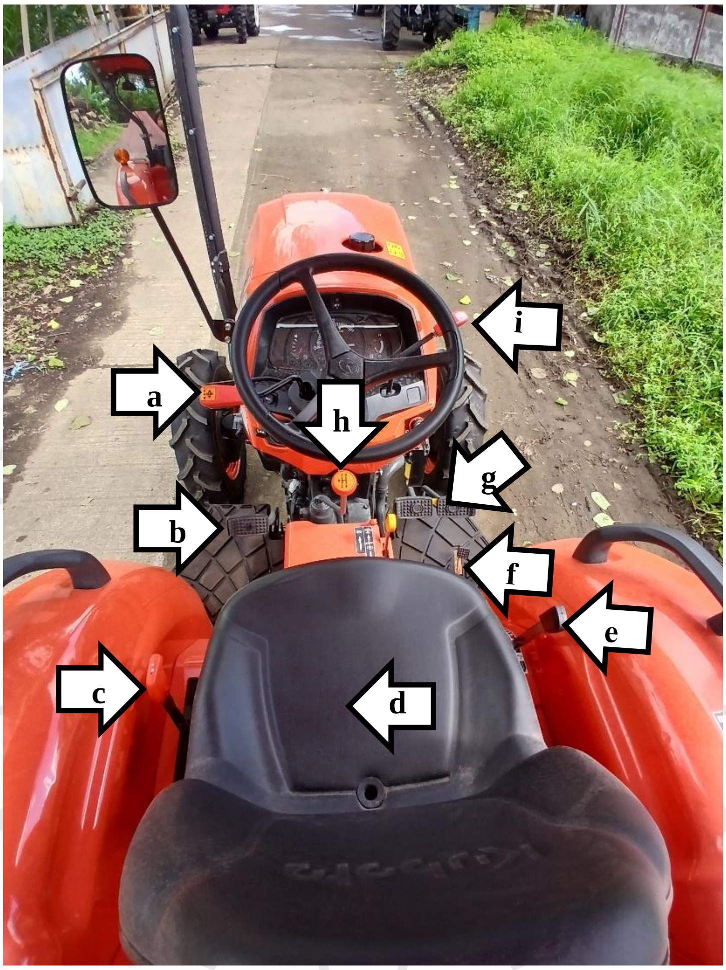

Figure 2. Controls of the KUBOTA L3228DT Four-Wheel Tractor include the following:
(a) Shuttle gear shift lever, (b) Main clutch pedal, (c) Range gear shift lever,
(d) Operator's seat, (e) Position control lever, (f) Foot throttle pedal,
(g) Brake pedals, (h) Main gear lever, and (i) Hand throttle lever.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Overall dimensions and weight of tractor		
1.1	Length, mm	2920	3273
1.2	Width, mm	1337	1345
1.3	Height, mm	2010	2150
1.4	Dry weight of tractor, kg	1257	1257[a]
2	Engine		
2.1	Brand	KUBOTA	KUBOTA
2.2	Make	ND	ND
2.3	Model	D1803-M-DI-E-TS6T	D1803-M-DI-E-TS6T[a]
2.4	Serial number	BRG1909	BRG1909[a]
2.5	Output power, kW	23.50	23.50[a]
2.6	Rated shaft speed, rpm	2700	2700[a]
2.7	Number of cylinders	3	3[a]
2.8	Bore × stroke, mm	87 × 102.4	87 × 102.4[a]
2.9	Displacement, cm^3	1.826	1.826[a]
2.10	Type	Direct injection, Vertical, Water-cooled 4 cycle diesel	Four-stroke cycle, three-cylinder, water-cooled, naturally-aspirated
2.11	Fuel used	Diesel	Diesel
2.12	Governor	ND	ND[b]
2.13	Air cleaner	ND	Dry-type, dual element
2.14	Lubrication system	ND	ND[b]
2.15	Cooling system	ND	Water-cooled
2.16	Starting system	ND	Electric start
2.17	Electrical system	ND	Battery
2.18	Exhaust system	ND	Vertical muffler
3	Steering system	Integral type power steering	Full hydrostatic

[a] The data were obtained from the operator's manual and nameplate provided by the test applicant.
[b] The engine was not disassembled.
Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
4	Power take-off		
4.1	Location	Rear	Rear of the tractor
4.2	Type	ND	Type I
4.2.1	Diameter of PTO shaft ends, mm	ND	34.69
4.2.2	Number of splines	6	6
4.2.3	Rated shaft speed, rpm	540, 750	540, 750
4.3	Height above ground, mm	ND	572
4.4	Direction of rotation (viewed from the rear of the tractor)	Clockwise	Clockwise
4.5	Mode of operation		
4.5.1	Speed change	ND	Lever-operated
4.5.2	Clutch	ND	Dependent
4.6	Rated PTO Power, kW	20.4	21.32[b]
5	Ground clearance, mm	330	330
6	Three-point linkage		
6.1	Category		
6.1.1	Based on PTO power at rated rotational frequency of engine	ND	1N, 1
6.1.2	Based on dimensions of tractor linkage point	ND	1N/1
6.2	Length of lift arms, mm	ND	225
6.3	Length of lower links, mm	ND	660
6.4	Horizontal distance between the two lower links, mm	ND	215
6.5	Horizontal distance between the two lift arms endpoints, mm	ND	330
6.6	Length of upper link, mm		
6.6.1	Minimum	ND	540
6.6.2	Maximum	ND	720
6.7	Distance of lower link pivot point to lift rod pivot points on lower links, mm	ND	290

[a] The data were obtained from the operator's manual provided by the requesting party.
[b] Based on actual PTO load test.
Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
6.8	Length of lift rods, mm		
6.8.1	Minimum	ND	405
6.8.2	Maximum	ND	495
6.9	Diameter of top hitch pin holes, mm	ND	19.36
6.10	Diameter of lower hitch pin holes, mm	ND	22.23
7	Drawbar		
7.1	Type	ND	Fixed
7.2	Category based on PTO power	ND	0, 1, 2, 3, 4, 5
7.3	Drawbar width, mm	ND	65.39
7.4	Drawbar thickness, mm		
7.4.1	Upper clevis	ND	16.31
7.4.2	Lower clevis	ND	16.12
7.5	Pin hole diameter, mm	ND	25.73
7.6	Pin diameter, mm	ND	25.4
7.7	F[b], mm	ND	32.34
7.8	G[b], mm	ND	145
7.9	Height, H, mm	ND	62.32
7.10	Throat depth, J, mm	ND	119.6
7.11	End radius of drawbar and clevis, R[b], mm	ND	48.04
7.12	W[b], mm	ND	NA

[a] The data were obtained from the operator's manual provided by the test applicant.
[b] Based on Table 6 of PNS/BAFS PABES 301:2020.
Note: Not Applicable (The edge of the drawbar is not chamfered), ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
8	Transmission system		
8.1	Main clutch and PTO clutch	Dry type single stage	Dependent
8.2	Transmission gears		
8.2.1	Type	Synchro-shuttle	Synchromesh
8.2.2	Main gear	1,2,3,4, N	1, 2, 3, 4, N
8.2.3	Range gear	High, Low, Neutral	High, Low, Neutral
8.2.4	Shuttle gear	Forward, Reverse, Neutral	Forward, Reverse, Neutral
8.3	Differential lock	Pedal	Pedal
9	Tire		
9.1	Type	Bias ply	Bias ply
9.2	Front tire size, W × D, mm	177.80 × 406.40 (7" × 16")	177.80 × 406.40 (7" × 16") [a]
9.3	Rear tire size, W × D, mm	284.48 × 609.60 (11.2" × 24")	284.48 × 609.60 (11.2" × 24") [a]
9.4	Wheelbase, mm	1610	1650
9.5	Wheel tread		
9.5.1	Front tread, mm	1020	1250
9.5.2	Rear tread, mm	1060	1335
10	Ballast		
10.1	Front ballast		
10.1.1	Number of ballasts	ND	5
10.1.2	Weight per ballast, kg	20	20
10.2	Wheel ballast	NA	NA
11	Brake system		
11.1	Type according to manner of applying braking force	ND	Mechanical
11.2	Type according to manner of transmitting the force from the controls	ND	Hydraulic
11.3	Parking brake	Lever-operated	Lever-operated

[a] The data were obtained from the component's markings and operator's manual provided by the test applicant.
Note: Not Applicable (No wheel ballast installed.), ND – No Data

RESULTS

Table 1. Radius of turning circle/area of the KUBOTA L3228DT Four-Wheel Tractor.

Item	Without Brake		With Brake	
	Right Turn	Left Turn	Right Turn	Left Turn
Radius of Turning Circle, mm	3330	3295	2370	2358
Radius of Turning Area, mm	3418	3383	2458	2445

Table 2. Traveling speed of the KUBOTA L3228DT Four-Wheel Tractor at different gear settings.

Range Gear Settings	Main Gear Settings	Shuttle Gear Settings	
		Forward (kph)	Reverse (kph)
Low	First	1.83	1.91
	Second	2.34	2.57
	Third	4.48	4.96
	Fourth	5.74	6.55
High	First	6.27	7.05
	Second	7.96	9.34
	Third	15.92	16.77
	Fourth	20.19	22.09

RESULTS (Cont'n)

Table 3. Test conditions during the power take-off load test of the KUBOTA L3228DT Four-Wheel Tractor.

Temperature, °C	33.7
Relative Humidity, %	63.4

Table 4. Power Take-Off Load Test data of the KUBOTA L3228DT Four-Wheel Tractor.

Speed		PTO		Fuel Consumption, L/h	Specific Fuel Consumption, g/kW-h
Engine, rpm	PTO, Rpm	Torque, N-m	Power, kW		
2906	642	55.50	3.73	3.30	743.93
2811	620	215.17	13.96	5.18	311.35
2720	600	339.83	21.32	6.75	265.72
2629	580	351.17	21.30	6.57	258.85
2538	560	356.00	20.85	6.31	254.19
2451	540	361.50	20.43	6.07	249.57
2365	520	369.50	20.11	5.91	246.67
2270	500	379.67	19.87	5.78	244.17
2179	480	390.17	19.61	5.61	240.13
2093	460	400.83	19.30	5.48	238.32
2005	440	409.83	18.88	5.32	236.47
1900	419	418.17	18.33	5.09	233.03

RESULTS (Cont'n)

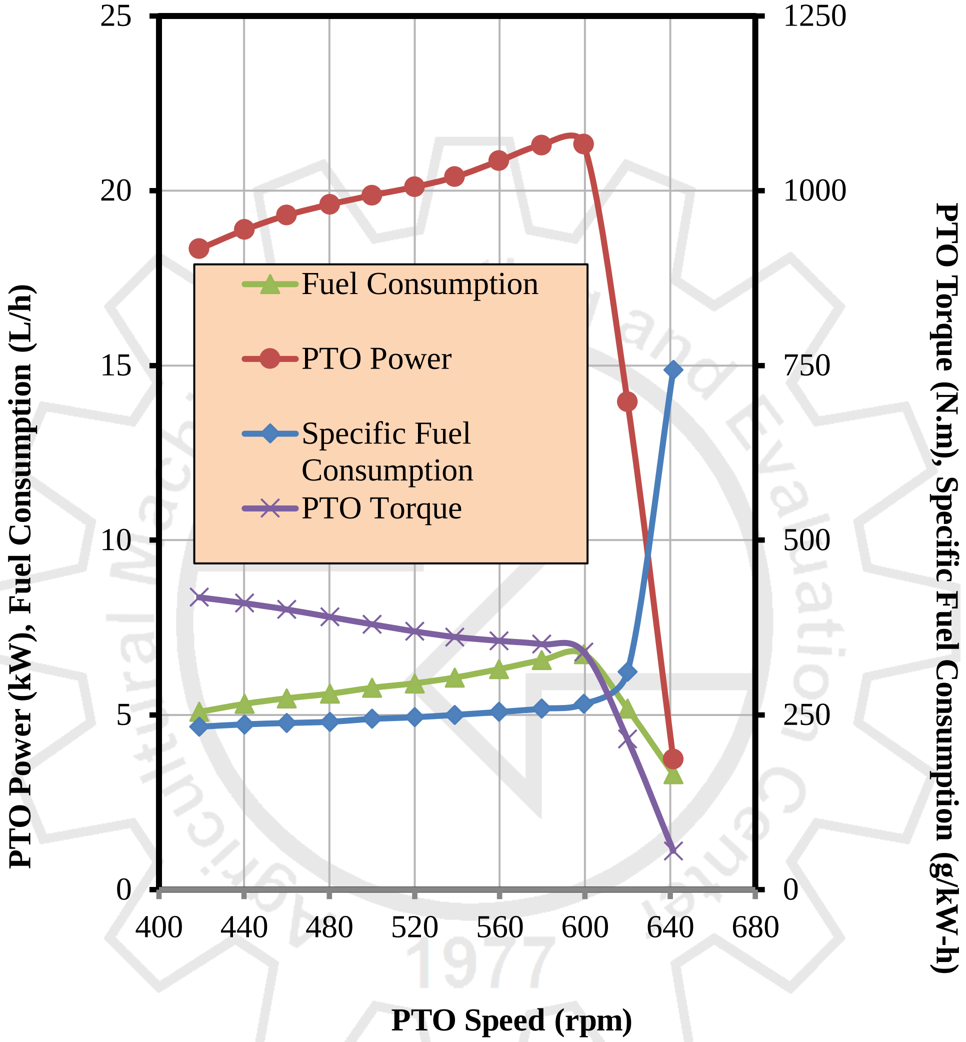

Figure 3. Performance Characteristics Curves of the KUBOTA L3228DT Four-Wheel Tractor.

RESULTS (Cont'n)

Table 5. Hydraulic Lift Test data of the KUBOTA L3228DT Four-Wheel Tractor without booster cylinder.

Trial	On the Frame	
	kg	kN
1	640.2	6.28
2	630.2	6.18
3	638.0	6.26
4	645.9	6.33
5	643.2	6.31
Average	**639.5**	**6.27**

Note: On the frame 610 mm from lower link end.

Table 6. Drawbar power and lift force capacity of the KUBOTA L3228DT Four-Wheel Tractor.

Drawbar power, kW	18.34[a]
Hydraulic lift force capacity, kN	5.68[b]

[a] 86% of maximum PTO power.

[b] Computed as per Table 8 of PNS/BAFS/PABES 301:2020, requirement if drawbar power is equal to 65 kW or below.

Figure 4. Hydraulic lift setup of the KUBOTA L3228DT Four-Wheel Tractor.

OBSERVATIONS

Figure 5. Nameplate of the KUBOTA L3228DT Four-Wheel Tractor.

Figure 6. Warning notices on the KUBOTA L3228DT Four-Wheel Tractor.

OBSERVATIONS (Cont'n)

Figure 7. Markings and labels on the KUBOTA L3228DT Four-Wheel Tractor.

OBSERVATIONS (Cont'n)

Figure 8. SMV-emblem provided on the tractor.

Tested by:

JOHN PAUL A. PALILLO
Test Engineer

Verified by:

MARIE JEHOSA B. REYES
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO	0 5 FEB 2025
AMTEC Director	Date

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